

Ad LLM as a Service

An LLM-Powered Conversational Advertising
Platform

A Project Report

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Abstract

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1 Introduction

1.1 Background and Motivation

1.2 Project Objectives

1.3 Scope and Constraints

1.4 Report Organisation

2 Literature Review

2.1 Evolution of Digital Advertising

2.2 Large Language Models and Conversational AI

2.3 Retrieval-Augmented Generation (RAG)

2.4 Advertising in Conversational Interfaces

2.5 Prompt Engineering for Controlled Generation

2.6 User Engagement Metrics in Conversational Systems

2.7 Summary and Research Gap

3 Problem Statement

3.1 Limitations of Traditional Ad Formats in Conversational AI

3.2 Technical Challenges

3.2.1 Real-Time Intent Extraction from Multi-Turn Conversations

3.2.2 Semantic Ad Matching at Low Latency

3.2.3 Natural Ad Integration Without Degrading User Experience

3.2.4 Engagement Measurement Beyond Clicks

3.3 Ethical and Regulatory Considerations

3.4 Formal Problem Definition

4 Solution Design

4.1 System Architecture Overview



Table 1: Technology stack summary.

Component	Technology
Frontend	React
Orchestrator API	FastAPI (Python)
Session Store	Redis
Database	PostgreSQL + pgvector
Embedding Model	text-embedding-3-small (OpenAI)
Context Extraction LLM	Claude Haiku (lightweight)
Response Generation LLM	Claude Sonnet (capable)
Event Logging	PostgreSQL (JSONB payload)

4.2 Component Design

4.2.1 Orchestrator API (FastAPI)

4.2.2 Context Extraction Module

4.2.3 Session Management (Redis)

4.2.4 Ad Inventory Store (PostgreSQL + pgvector)

4.2.5 Embedding and Semantic Matching Pipeline

4.2.6 Re-Ranking and Business Logic

Algorithm 1 Ad Candidate Re-Ranking

Require: Query embedding $\mathbf{q}$, context C , session ID s

Ensure: Ranked list of top- k ad candidates

- 1: $raw \leftarrow \text{VectorSearch}(\mathbf{q}, \text{limit} = 20)$ $\triangleright$ pgvector cosine similarity
 - 2: $shown \leftarrow \text{Redis.get}(s).\text{ads_shown}$
 - 3: $candidates \leftarrow \{a \in raw \mid a.\text{active} \wedge a.\text{budget} > 0 \wedge a.\text{id} \notin shown\}$
 - 4: $b_{\max} \leftarrow \max_{a \in candidates} a.\text{bid_cpm}$
 - 5: **for all** $a \in candidates$ **do**
 - 6: $a.\text{score} \leftarrow 0.5 \cdot a.\text{sim} + 0.3 \cdot \frac{a.\text{bid_cpm}}{b_{\max}} + 0.2 \cdot C.\text{purchase_signal}$
 - 7: **end for**
 - 8: **return** $\text{TopK}(candidates, k = 3)$
-

4.2.7 Response Synthesis with Ad Injection

4.2.8 Click Tracking and Event Logging

4.2.9 Analytics Dashboard

4.3 Interface Contracts and Workstream Division

Table 2: Interface contracts between Backend and AI workstreams.

Function	Caller → Callee	Description
<code>extract_context</code>	Backend → AI	Extract structured context from user message and history
<code>embed_query</code>	Backend → AI	Generate query embedding from extracted context
<code>embed_ad</code>	Backend → AI	Generate ad embedding at ingestion time
<code>get_candidate_ads</code>	AI → Backend	Retrieve and rank ad candidates via vector search
<code>generate_response</code>	Backend → AI	Synthesise response with optional ad injection

4.4 Quality Assurance and Evaluation Framework

4.5 Error Handling and Graceful Degradation

5 Illustrative Examples

5.1 Example 1: Product Research Conversation

5.2 Example 2: Sensitive Conversation (Ad Suppression)

5.3 Example 3: Irrelevant Ad Rejection

5.4 Example 4: Engagement Follow-Up

5.5 Example 5: Multi-Turn Context Accumulation

6 Conclusions

6.1 Summary of Contributions

6.2 Workstream Contributions

6.3 Limitations

6.4 Future Work

6.5 Reflections

A Database Schema**B API Endpoint Specifications****C Prompt Templates****D Context Extraction Test Cases****E Response Synthesis Evaluation Set****F Session State Schema****G Project Timeline**

Table 3: Project timeline by workstream.

Week	Backend Workstream	AI/LLM Workstream	Milestone
1	Orchestrator skeleton, Redis, Postgres + pgvector schema	Context extraction prompt design + module	Infrastructure foundation
2	Ad CRUD API, bulk ingestion, re-ranking logic	Embedding functions, retrieval quality testing	Ad matching pipeline ready
3	Click tracking, event logging, tracking link generator	Response synthesis prompt engineering + module	Core pipeline complete
4	Analytics API, frontend integration, end-to-end wiring	Eval set, quality tuning, engagement classifier	End-to-end integration
5	Error handling, monitoring, hardening	LLM-as-judge evaluator, attention model start	Demo-ready MVP

